Simulation
IEORE4404
Fall 2025
Calvin Chang Homework 3

Problem 1.

Solution: Given the Cumulative Distribution Function for L, we can compute the inverse function G(u) as:

$$U = 1 - \frac{1}{x^3} \tag{1}$$

$$\frac{1}{x^3} = 1 - U \tag{2}$$

$$x^3 = \frac{1}{1 - U} \tag{3}$$

$$x = \sqrt[3]{\frac{1}{1 - U}} \tag{4}$$

We can then use this to generate X using the Unif(0,1). The pseudo-code of which would be:

```
generate u = Unif(0,1)
let x = (1 - u)^(1/3)
output x
```

This code can then be written in python to create the distribution:

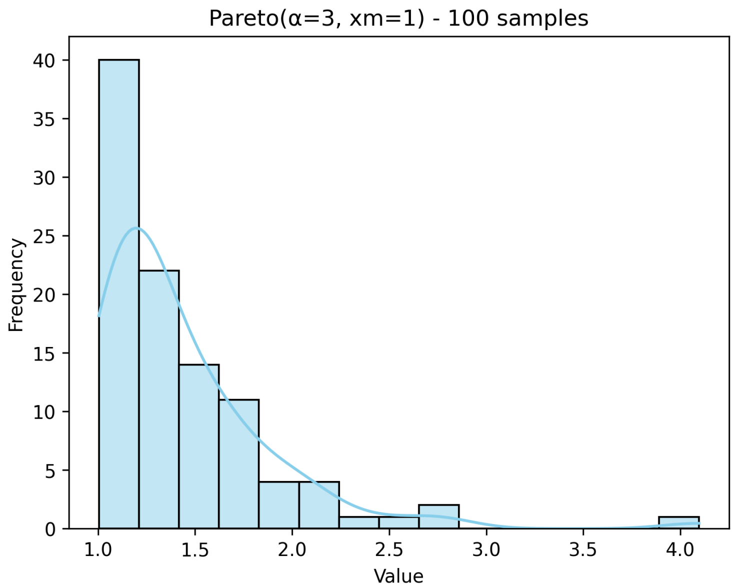

Figure 1: Distribution with 100 simulated samples

Problem 2.

Solution:

a) **Inverse transform method:** Given the density function, we can find the Cumulative Distribution Function by:

$$\begin{aligned}\int_0^x \frac{e^t}{e-1}\,dt &= \frac{1}{e-1}\int_0^x e^t\,dt &(5)\\ &= \frac{1}{e-1}\left(e^x - e^0\right) &(6)\\ &= \frac{e^x-1}{e-1}. &(7)\end{aligned}$$

Given $F(x)$, we can now use Unif$(0,1)$ to generate this random variable:

$$\begin{aligned}U &= \frac{e^x-1}{e-1} &(8)\\ U(e-1)+1 &= e^x &(9)\\ x &= \ln\big(U(e-1)+1\big). &(10)\end{aligned}$$

Therefore, the inverse function is

$$G(u) = \ln\big(U(e-1)+1\big).$$

We can then use this to generate x using the Unif(0,1). The pseudo-code of which would be:

```
generate u = Unif(0,1)
let x = ln(U*(e-1)+1)
```

b) **Acceptance-rejection method:** Given the density function, we can use a uniform proposal $g(x) = 1, 0 \leq x \leq 1$.

$$\begin{aligned}\frac{f(x)}{g(x)} &= \frac{e^x}{e-1} \leq \frac{e}{e-1} &(11)\\ C &= \frac{e}{e-1} &(12)\end{aligned}$$

To generate X, we can use the pseudo-code:

```
generate Y = Unif(0,1)
generate U = Unif(0,1)
if U <= f(Y)/C*g(Y):
    X = Y (accept)
else:
    (reject)
output X
```

c) My preferred method is the Inverse transform method. Mainly because the method only has to use one of the Uniform(0,1) functions, making the process 100% efficient and all outputs are accepted when running code. On the other hand, the acceptance-rejection method will only have a $\frac{1}{C} = \frac{(e-1}{e} \approx 0.632$ acceptance rate.

d) plots below:

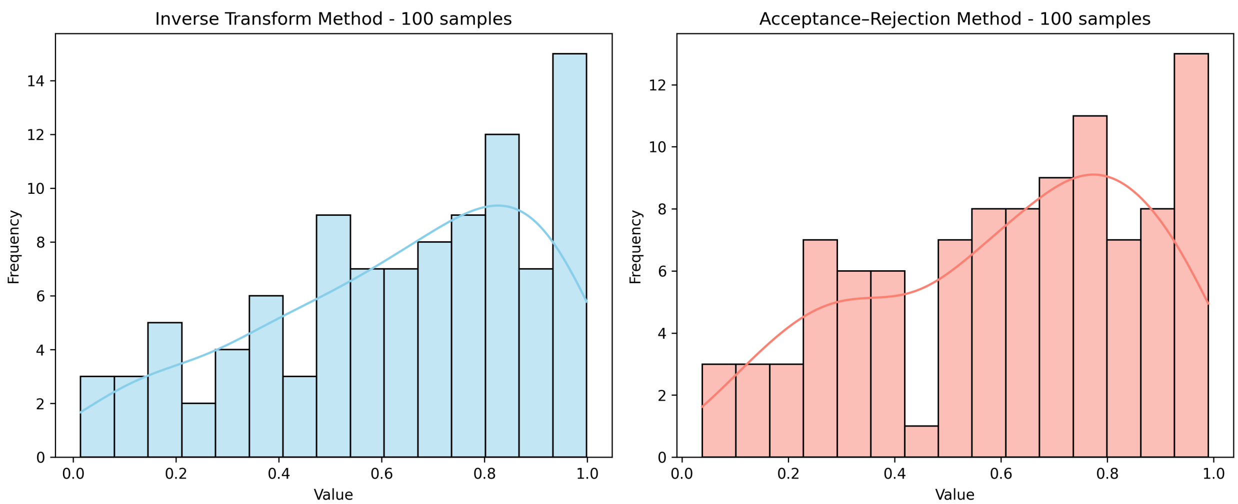

Figure 2: Comparison of distribution with 100 simulated samples

Problem 3.

Solution:

a) Given the gamma distribution:

$$\frac{f(x)}{g(x)} = \frac{\frac{e^x x^{n-1}}{(n-1)!}}{\frac{1}{2}e^{\frac{-x}{2}}} \tag{13}$$

$$= \frac{2}{(n-1)!}x^{n-1}e^{-x/2} \tag{14}$$

Then, to find the C, we have to find max of $\frac{f(x)}{g(x)}$:

$$h(x) = x^{n-1}e^{-x/2} \tag{15}$$

$$ln(h(x)) = (n-1)lnx - x/2 \tag{16}$$

$$\frac{d}{dx}ln(h(x)) = (n-1)\frac{1}{x} - \frac{1}{2} \tag{17}$$

$$\tag{18}$$

We can then let $(n-1)\frac{1}{x} - \frac{1}{2} = 0$, which then gives:

$$x = 2(n-1) \tag{19}$$

$$\frac{d^2}{dx^2}ln(h(x)) = -(n-1)\frac{1}{x^2} < 0 \tag{20}$$

Therefore when $x = 2(n-1)$, $h(x)$ is at the max, and we can therefore conclude that

$$C = \frac{2}{(n-1)!}[2(n-1)]^{n-1}e^{-(n-1)} \tag{21}$$

To generate X, the following pseudo code can be used:

```
generate U1 = uniform(0,1), Y = -2*ln(U1) = Exp(delta = 1/2)
generate U2 = uniform(0,1)
if U <= f(Y)/C*g(Y) -> [Y^(n-1)*e^(-Y/2)]/[(2(n-1))^(n-1)*e^(-(n-1))
    ]:
    set X = Y (accept)
else:
    (reject)
output X
```

b) Using an exponential density would **not** work, this is because if

$$g(x) = e^{-x} \tag{22}$$

$$\frac{f(x)}{g(x)} = \frac{x^{n-1}}{(n-1)!} \tag{23}$$

For $n \geq 2$, this ratio will increase to infinity as $x \to \infty$, which means that there is no finite constant C that satisfies condition $f(x) \leq Cg(x)$ for all $x \geq 0$. This goes against the acceptance-rejection requirement, since the proposal's tail decays too fast. The only case where Exp(1) would work is when $n = 1$, in which case $f(x) = g(x), C = 1$

Problem 4.

Solution:

a) Given the semi-circular distribution:

$$g(x) = Unif(-1, 1) \tag{24}$$
$$= \frac{1}{2}, -1 \leq x \leq 1 \tag{25}$$

We can then find:

$$\frac{f(x)}{g(x)} = \frac{2/\pi\sqrt{1-x^2}}{1/2} \tag{26}$$
$$= 4/\pi\sqrt{1-x^2} \tag{27}$$

given $4/\pi\sqrt{1-x^2} \leq 4/\pi$, we can find $C = 4/\pi$. We can then use the following pseudo-code to generate X:

```
generate U1 = uniform(0,1)
set Y = -1 + 2U, Y = uniform(-1,1)
generate U = uniform(0,1)
if U <= f(Y)/C*g(Y) -> sqrt(1-Y^2):
    set X = Y (accept)
else:
    (reject)
output X
```

b) The acceptance probability:

$$p = \frac{1}{C} = \frac{\pi}{4} \approx 0.785 \tag{28}$$

The expected number of trials required to get an output would then be:

$$\frac{1}{p} = \frac{4}{\pi} \approx 1.27 \tag{29}$$